

Technical Note: Task-Visible Operator Mechanisms for In-Context Linear Ridge Regression

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1 Introduction

The goal is to test whether a trained transformer implements the linear-kernel ridge-regression operator, not just whether it predicts one label vector correctly. This distinction matters because many different algorithms can agree on one sampled label vector while responding differently to nearby label perturbations. We therefore turn each episode into an operator test: freeze the inputs and ask how predictions change as the context labels are varied.

Fix one episode and hold all non-label inputs fixed:

$$G = (X_{\text{ctx}}, X_{\text{tgt}}).$$

Let

$$X_{\text{ctx}} \in \mathbb{R}^{n_{\text{ctx}} \times d_x}, \quad X_{\text{tgt}} \in \mathbb{R}^{n_{\text{tgt}} \times d_x}.$$

The transformer receives the fixed sequence

$$(x_1, y_1, 0), \dots, (x_{n_{\text{ctx}}}, y_{n_{\text{ctx}}}, 0), (x_1^*, 0, 1), \dots, (x_{n_{\text{tgt}}}^*, 0, 1),$$

where the first block contains context tokens and the second block contains target tokens. In the operator tests, only the context-label entries $y_1, \dots, y_{n_{\text{ctx}}}$ are perturbed. Rows of X_{ctx} and X_{tgt} are sampled from a Gaussian input law. Conditional on an episode covariance Σ ,

$$x_i \sim \mathcal{N}(0, \Sigma), \quad \Sigma = U\Lambda U^\top,$$

where U is a random orthogonal matrix obtained by QR-orthogonalizing a Gaussian matrix. During training, Λ is sampled from the spectral curriculum described in the experimental setup. The operator experiments reported below instead use the flat-spectrum evaluation geometry

$$\Sigma = cI, \quad c \sim \text{Unif}[1, 10],$$

implemented by the shared flat eigenvalues sampler. Set

$$K := X_{\text{ctx}}X_{\text{ctx}}^\top, \quad K_t := X_{\text{tgt}}X_{\text{ctx}}^\top, \quad A := K + \sigma^2 I_{n_{\text{ctx}}}, \quad \sigma^2 > 0.$$

These matrices are introduced because, once the inputs are fixed, all of ridge regression is encoded by the context Gram matrix K , the target-context Gram matrix K_t , and the regularized system matrix A . Labels are generated by a sampled teacher,

$$\beta \sim \mathcal{N}(0, I_{d_x}), \quad y = X_{\text{ctx}}\beta + \eta, \quad \eta \sim \mathcal{N}(0, \sigma^2 I_{n_{\text{ctx}}}).$$

The target values used for evaluation are the noiseless teacher values $X_{\text{tgt}}\beta$. This is the same sampling convention used by the experiment scripts: context labels are noisy, target labels are not given to the transformer, and KRR is evaluated as the conditional predictor from context labels to target values. Ridge regression does not observe β . It forms its estimate from the sampled inputs and context labels. For context labels $y \in \mathbb{R}^{n_{\text{ctx}}}$, exact linear-kernel ridge regression gives

$$\alpha^* = A^{-1}y, \quad y^* = K_t \alpha^*.$$

Equivalently, after the geometry is fixed, ridge regression is the conditional operator

$$T_G := K_t A^{-1}, \quad y^* = T_G y.$$

Equivalently, in primal form,

$$\hat{\beta} = \left(X_{\text{ctx}}^\top X_{\text{ctx}} + \sigma^2 I_{d_x} \right)^{-1} X_{\text{ctx}}^\top y, \quad y^* = X_{\text{tgt}} \hat{\beta}.$$

Every behavioral, representational, and causal test below is ultimately scored against this operator, not only against the realized prediction $T_G y$.

Let the trained transformer define the fixed-geometry label map

$$F_\theta^G : \mathbb{R}^{n_{\text{ctx}}} \rightarrow \mathbb{R}^{n_{\text{tgt}}}, \quad F_\theta^G(y) = \hat{y}_\theta.$$

When the geometry is clear, we write F and T for F_θ^G and T_G . The central comparison is therefore between two maps with the same input and output spaces: the transformer’s local label-response map F and the exact KRR operator T .

A context-label direction is a vector $u \in \mathbb{R}^{n_{\text{ctx}}}$ added to the context labels while the input geometry G is held fixed. Testing a direction means comparing the transformer’s response to $y + \varepsilon u$ and $y - \varepsilon u$. Thus the word “direction” always refers to a direction in label space over the context positions, not to a direction in input-feature space or target space.

The transformer’s activations are the hidden states produced by the trained transformer on the same fixed episode. When we speak about an activation subspace, we mean a subspace over context positions extracted from context-token hidden states. Later sections define how these subspaces are tested against the ridge operator.

This note separates the mechanism claim into three tests.

1. **Behavior.** Does the transformer’s local response to label changes match T on held-out perturbations?
2. **Representation.** Do the transformer’s activations contain a compact context-side subspace from which the same operator can be recovered?
3. **Causal use.** Once the relevant activation directions have been identified, does removing them damage the transformer’s response?

The next section defines the rank requirement before inspecting the transformer. It is a rank result, not a learning guarantee. The experiments then check whether the trained transformer exposes a subspace meeting that requirement, whether the transformer response agrees with the operator built from that subspace, and whether the transformer causally relies on the extracted directions. The claim is local to task-distribution label directions; isotropic probes show that the transformer’s learned response drifts off distribution.

All theoretical claims are for the linear kernel. The studied transformer is bidirectional, so the explanation is for the conditional prediction operator $K_t A^{-1}$, not for a target-independent stored dual solution.

High-level construction. The construction has one organizing idea. After the input geometry is fixed, ridge regression is a linear map from context labels to target predictions. We want to know whether the transformer realizes this map through a small context-side activation subspace. The plan is:

1. First define the ideal subspace problem without using the transformer: what is the smallest context-label subspace that can reproduce the ridge operator on task-distributed label directions?
2. Then extract candidate subspaces from the trained transformer’s activations and test whether they are sufficient for the same ridge operator.
3. Finally test three things separately: whether the transformer response matches ridge regression, whether the extracted subspace is sufficient for ridge regression, and whether the transformer causally depends on the extracted directions.

Thus the main claim is: the transformer contains a compact activation subspace on the scale of the rank requirement defined below for the ridge operator; the operator built from that subspace is accurate; and removing the relevant extracted directions damages the transformer’s task-local response.

Norms. The A -norm is the geometry induced by the ridge linear system. For $v \in \mathbb{R}^{n_{\text{ctx}}}$ and $M \in \mathbb{R}^{n_{\text{ctx}} \times d}$, define

$$\|v\|_A := \sqrt{v^\top A v}, \quad \|M\|_{A,F} := \|A^{1/2} M\|_F.$$

All prediction-space norms are Euclidean or Frobenius norms. A fixed numerical floor $\eta_{\text{num}} > 0$ appears only in denominators; in experiments $\eta_{\text{num}} = 10^{-12}$.

What is not claimed. The pre-transformer theory supplies a rank requirement. It does not say that raw depth, head count, or width imply ridge behavior. It also does not give an SGD or training guarantee. The layerwise activation-span diagnostic is a post-training diagnostic of where the relevant directions appear, not a pre-training architecture budget.

2 Task-Visible Rank

This section gives the only pre-transformer theorem used in the paper. It depends only on the input distribution, the label law, and the desired accuracy. Its purpose is to define the target that any activation explanation must meet. Before looking inside the transformer, we ask: how many context-label directions are needed, in principle, to reproduce ridge predictions on the label directions the task actually produces? This number is the benchmark for the later activation construction. If the transformer exposes fewer useful directions than this, the proposed subspace cannot approximate the ridge operator on the task distribution; if it exposes enough, the remaining question is whether the transformer’s local response is close to the corresponding subspace-based approximation and whether the transformer causally depends on that subspace.

Assume the matched random linear regression task

$$y = X_{\text{ctx}} \beta + \eta, \quad \beta \sim \mathcal{N}(0, I_{d_x}), \quad \eta \sim \mathcal{N}(0, \sigma^2 I_{n_{\text{ctx}}}).$$

Here the randomness in X_{ctx} and X_{tgt} first samples the geometry G ; the randomness in β and η then samples labels conditional on that geometry. Conditional on the geometry G ,

$$\text{Cov}(y \mid G) = A.$$

Thus task-label directions are isotropic in the whitened coordinates $A^{-1/2}y$. Define the task-visible operator

$$C_G := K_t A^{-1/2}.$$

Let $s_1(G) \geq s_2(G) \geq \dots \geq 0$ be its singular values. The whitening is the reason C_G is the right object. In coordinates where task labels are standard Gaussian, C_G is exactly the ridge map from whitened label directions to target predictions. Its singular spectrum therefore says which task-label directions matter most for prediction.

Definition 1 (Task-visible rank). *For fixed G and tolerance $\epsilon > 0$, define*

$$r_{\text{task}}(G; \epsilon) := \min \left\{ r : \frac{\sum_{i>r} s_i(G)^2}{\sum_i s_i(G)^2 + \eta_{\text{num}}} \leq \epsilon^2 \right\}.$$

For random geometries $G \sim \mathcal{D}_G$, define

$$r_{\text{task}}^{\text{hp}}(\epsilon, \delta) := \min \{ r : \mathbb{P}_G(r_{\text{task}}(G; \epsilon) \leq r) \geq 1 - \delta \}.$$

This rank is the number of target-visible ridge directions needed on the task label distribution. It is a requirement that an activation subspace may or may not meet after training; it is not a raw architectural budget. The definition is intentionally an effective rank rather than the algebraic rank of T . Algebraic rank counts every nonzero singular direction, even one with negligible prediction energy. By contrast, $r_{\text{task}}(G; \epsilon)$ asks for the smallest number of directions whose omission leaves at most an ϵ^2 fraction of task-visible ridge-response energy unexplained. In the experiments below this distinction is what makes r_{task} the principled target for $\dim \text{span}(Q)$: if the extracted Q -space has dimension below this value, no aligned reduced ridge operator of that dimension can reach the stated tolerance on task probes.

For the linear kernel,

$$\text{rank}(C_G) = \text{rank}(T) = \text{rank}(K_t) \leq \min\{d_x, n_{\text{ctx}}, n_{\text{tgt}}\},$$

because A is invertible and $K_t = X_{\text{tgt}} X_{\text{ctx}}^\top$. Thus

$$r_{\text{task}}(G; \epsilon) \leq \text{rank}(T) \leq \min\{d_x, n_{\text{ctx}}, n_{\text{tgt}}\}.$$

In the flat-spectrum random geometries used in the target-rank sweep, this upper bound is usually attained at the reported tolerance. To see why, write a thin SVD $X_{\text{ctx}} = P \Sigma V^\top$. Then

$$C_G = K_t A^{-1/2} = X_{\text{tgt}} V \text{diag} \left(\frac{\sigma_i}{\sqrt{\sigma_i^2 + \sigma^2}} \right) P^\top.$$

With $n_{\text{ctx}} = 47$ and $\sigma^2 = 0.1$, the context singular values in the flat random linear task are typically much larger than the ridge noise. For an isotropic Gaussian $n_{\text{ctx}} \times d_x$ context matrix with feature variance c , the squared singular values concentrate on the scale $c(\sqrt{n_{\text{ctx}}} \pm \sqrt{d_x})^2$, so the shrinkage factors $\sigma_i / \sqrt{\sigma_i^2 + \sigma^2}$ are close to one in the reported $d_x \leq 15$ sweep. The remaining factor $X_{\text{tgt}} V$

is a random $n_{\text{tgt}} \times d_x$ target-side matrix; when $d_x \leq n_{\text{tgt}} = 16$, it is generically full column rank and its nonzero singular directions are not small enough to fall below the tail-energy tolerance. Hence the empirical task-visible rank is expected to be $r_{\text{task}} \approx d_x$ for the main linear sweep. For the additional $d_x = 30$ stress row, $n_{\text{tgt}} = 16$ caps the task-visible rank.

The next quantity turns the rank question into an operator approximation question. Suppose two candidate label-to-prediction maps, S_1 and S_2 , are equal on one realized label vector but differ on other possible label directions. We need a way to say how different they are on the directions we care about. A probe law P specifies those directions. For example, the task probe law samples directions with the same covariance as task labels, while the isotropic law samples all context-label directions equally.

For a fixed geometry and a probe law P , define the population operator error

$$E_P(S_1, S_2) := \left(\frac{\mathbb{E}_{u \sim P} \|(S_1 - S_2)u\|_2^2}{\mathbb{E}_{u \sim P} \|Tu\|_2^2 + \eta_{\text{num}}} \right)^{1/2}.$$

The expectation is only over the probe direction u ; the geometry is already fixed. The numerator is the mean squared prediction-response discrepancy between S_1 and S_2 on the chosen directions. The denominator is the mean squared response of the exact ridge operator on the same directions. Thus $E_P(S_1, S_2) = 0.01$ means that the root-mean-square operator discrepancy is about one percent of the root-mean-square ridge response under P . This normalization lets the theorem and the experiments use the same scale, and it prevents directions where ridge has almost no response from dominating the interpretation.

Theorem 1 (Task-visible rank theorem). *Fix a geometry G in the matched linear task. Let $T = K_t A^{-1}$ and let $Q \in \mathbb{R}^{n_{\text{ctx}} \times r}$ be A -orthonormal, $Q^\top A Q = I_r$. Define the reduced operator*

$$T_Q := K_t Q Q^\top.$$

Let P_{task} be the task-label probe law $u \mid G \sim \mathcal{N}(0, A)$. Then

$$\min_{Q^\top A Q = I_r} E_{P_{\text{task}}}(T_Q, T)^2 = \frac{\sum_{i > r} s_i(G)^2}{\sum_i s_i(G)^2 + \eta_{\text{num}}},$$

where $s_i(G)$ are the singular values of $C_G = K_t A^{-1/2}$. A minimizer is

$$Q_r^* = A^{-1/2} V_{1:r},$$

where $V_{1:r}$ are the top right singular vectors of C_G . Consequently, if $r \geq r_{\text{task}}(G; \epsilon)$, then there exists an r -dimensional reduced operator with

$$E_{P_{\text{task}}}(T_Q, T) \leq \epsilon.$$

If $r \geq r_{\text{task}}^{\text{hp}}(\epsilon, \delta)$, the same statement holds with probability at least $1 - \delta$ over sampled geometries.

Proof. Let $W = A^{1/2} Q$. Then $W^\top W = I_r$ and

$$T_Q = K_t A^{-1/2} W W^\top A^{-1/2}.$$

Write $u = A^{1/2} z$. Under the task-label probe law, $z \sim \mathcal{N}(0, I)$. Therefore

$$Tu = C_G z, \quad T_Q u = C_G W W^\top z,$$

so

$$(T - T_Q)u = C_G(I - WW^\top)z.$$

Taking expectation over z gives

$$\mathbb{E}\|(T - T_Q)u\|_2^2 = \|C_G(I - WW^\top)\|_F^2, \quad \mathbb{E}\|Tu\|_2^2 = \|C_G\|_F^2.$$

The optimization is therefore

$$\min_{W^\top W = I_r} \|C_G(I - WW^\top)\|_F^2.$$

By Ky Fan’s maximum principle, this residual is minimized by the top right singular vectors of C_G , and the minimum residual is $\sum_{i>r} s_i^2$. The high-probability statement follows from the definition of $r_{\text{task}}^{\text{hp}}$. $\square$

Remark 1 (Meaning of the sufficient condition). *The theorem above is a rank theorem for context-side subspaces. It says that an A -orthonormal trial space of sufficient task-visible dimension can represent the ridge operator on task-label probes. It does not say that a transformer learns, exposes, or uses such a space. Those are post-training questions.*

This leaves the central empirical problem. The theorem tells us what a sufficient context-side subspace would have to accomplish, but it does not tell us whether the trained transformer has such a subspace or whether the transformer’s predictions depend on it. The next section therefore converts the rank condition into measurable diagnostics: first compare the transformer’s local label-response map to ridge regression, then compare activation-derived reduced operators to both the transformer and the exact ridge operator.

3 Operator Diagnostics and Activation Certificates

Section 2 defined the ideal trial-space target. This section explains how to test that target in a trained transformer. The central principle is to freeze the geometry, perturb only labels, and compare the resulting local response to the operator T . We use local perturbations because the transformer is nonlinear as a function of the whole token sequence, even if the target algorithm is linear in the labels after the geometry is fixed.

Definition 2 (Central finite-difference response). *For step size $\varepsilon > 0$ and direction $u \in \mathbb{R}^{n_{\text{ctx}}}$, define*

$$D_\varepsilon F(y)[u] := \frac{F(y + \varepsilon u) - F(y - \varepsilon u)}{2\varepsilon}.$$

This is the measured label-to-prediction response of the transformer in direction u . If the transformer locally implements ridge regression, this quantity should be close to Tu on held-out label directions.

The population error from Section 2 is the ideal quantity, but experiments only use finitely many label directions. The next definition is its empirical version. The first error compares a nonlinear transformer to a fixed linear operator by asking: when the labels are nudged in direction u_j , does the transformer’s measured response agree with what operator S would have produced on the same direction? Taking several held-out directions makes this an operator test rather than a test on the original label vector alone.

Definition 3 (Operator errors). *Let $U = \{u_1, \dots, u_m\}$ be a predeclared set of probe directions. For a fixed linear operator $S : \mathbb{R}^{n_{\text{ctx}}} \rightarrow \mathbb{R}^{n_{\text{tgt}}}$, define*

$$E_\varepsilon(F, S; U) := \left(\frac{\sum_{j=1}^m \|D_\varepsilon F(y)[u_j] - Su_j\|_2^2}{\sum_{j=1}^m \|Tu_j\|_2^2 + \eta_{\text{num}}} \right)^{1/2}.$$

For two linear operators S_1, S_2 , define

$$E(S_1, S_2; U) := \left(\frac{\sum_{j=1}^m \|(S_1 - S_2)u_j\|_2^2}{\sum_{j=1}^m \|Tu_j\|_2^2 + \eta_{\text{num}}} \right)^{1/2}.$$

The corresponding population quantity $E_P(S_1, S_2)$ was defined in Section 2.

Proposition 1 (Scale of the task-probe reduced-operator error). *Let P_{task} be the task-label probe law $u = A^{1/2}z$, $z \sim \mathcal{N}(0, I)$, and let $Q^\top A Q = I$. Set*

$$C = K_t A^{-1/2}, \quad W = A^{1/2} Q, \quad P_Q = W W^\top.$$

Then

$$E_{P_{\text{task}}}(T_Q, T)^2 = \frac{\|C(I - P_Q)\|_F^2}{\|C\|_F^2 + \eta_{\text{num}}},$$

and therefore

$$0 \leq E_{P_{\text{task}}}(T_Q, T) \leq 1.$$

If W is the span of the top r right singular vectors of C , then

$$E_{P_{\text{task}}}(T_Q, T)^2 = \frac{\sum_{i>r} s_i^2}{\sum_i s_i^2 + \eta_{\text{num}}},$$

where s_i are the singular values of C .

Proof. Since $W = A^{1/2}Q$, $W^\top W = Q^\top A Q = I$, so $P_Q = W W^\top$ is an orthogonal projection. For $u = A^{1/2}z$,

$$Tu = K_t A^{-1} A^{1/2} z = Cz,$$

while

$$T_Q u = K_t Q Q^\top A^{1/2} z = K_t A^{-1/2} W W^\top z = C P_Q z.$$

Thus $(T - T_Q)u = C(I - P_Q)z$. Taking expectation over $z \sim \mathcal{N}(0, I)$ gives

$$\mathbb{E}\|(T - T_Q)u\|_2^2 = \|C(I - P_Q)\|_F^2, \quad \mathbb{E}\|Tu\|_2^2 = \|C\|_F^2.$$

Because right multiplication by an orthogonal projection is a contraction in Frobenius norm, $\|C(I - P_Q)\|_F \leq \|C\|_F$, proving the bound. The singular-value formula follows from the Eckart–Young–Mirsky theorem, or equivalently Ky Fan’s maximum principle, applied to the best rank- r right singular subspace of C . $\square$

Proposition 2 (No universal upper bound for transformer-response errors). *Fix a finite probe set U with ridge-response energy*

$$R_U^2 = \sum_{j=1}^m \|Tu_j\|_2^2 > 0.$$

If the measured local response is zero on all probes, $D_\epsilon F(y)[u_j] = 0$, then

$$E_\epsilon(F, T; U) = \frac{R_U}{(R_U^2 + \eta_{\text{num}})^{1/2}} \leq 1.$$

If the measured response is the negative ridge response, $D_\epsilon F(y)[u_j] = -Tu_j$, then

$$E_\epsilon(F, T; U) = \frac{2R_U}{(R_U^2 + \eta_{\text{num}})^{1/2}} \leq 2.$$

In contrast, $E_\epsilon(F, T; U)$ has no universal upper bound.

Proof. The first two identities follow by substituting $D_\epsilon F(y)[u_j] = 0$ and $D_\epsilon F(y)[u_j] = -Tu_j$ into the definition. For unboundedness, suppose on the probe set the measured response is $D_\epsilon F(y)[u_j] = cTu_j$. Then

$$E_\epsilon(F, T; U) = |c - 1| \frac{R_U}{(R_U^2 + \eta_{\text{num}})^{1/2}},$$

which diverges as $|c| \rightarrow \infty$ whenever $R_U > 0$. $\square$

Proposition 3 (Triangle decomposition). *For any fixed linear operator S and any finite probe set U ,*

$$E_\epsilon(F, T; U) \leq E_\epsilon(F, S; U) + E(S, T; U).$$

In particular, taking $S = T_Q$ separates transformer-to-subspace error from subspace-to-ridge error.

Proof. Stack the probe responses as columns. With

$$M_F = [D_\epsilon F(y)[u_1] \ \cdots \ D_\epsilon F(y)[u_m]], \quad M_S = [Su_1 \ \cdots \ Su_m], \quad M_T = [Tu_1 \ \cdots \ Tu_m],$$

the numerator of each error is a Frobenius norm, and all three errors have the same denominator $(\|M_T\|_F^2 + \eta_{\text{num}})^{1/2}$. The ordinary triangle inequality gives

$$\|M_F - M_T\|_F \leq \|M_F - M_S\|_F + \|M_S - M_T\|_F,$$

and division by the common denominator proves the claim. $\square$

Remark 2 (How to read the error values). *For the population task-probe reduced-operator error, the scale is bounded between 0 and 1. An error e is a relative RMS response error, and e^2 is the fraction of task-probe ridge-response energy left unexplained by the reduced operator. Thus $e = 0.01$ means about one percent RMS error, $e = 0.1$ means about one percent squared response energy is missed, and $e = 0.54$ means roughly $0.54^2 \approx 29\%$ of the task response energy is missed. Values near 1 mean that the reduced operator leaves almost all task-visible ridge response unexplained.*

For transformer errors $E_\epsilon(F, T)$ and $E_\epsilon(F, T_Q)$, there is no analogous upper bound. A value near 1 is already comparable to a zero-local-response baseline, values above 1 are worse than having no ridge-like label response on the probes, and very large values indicate that the finite-difference response is badly misaligned or unstable relative to the ridge response. All these readings are probe-law dependent: task-probe and isotropic-probe errors answer different questions.

The first quantity, $E_\epsilon(F, S; U)$, is used whenever one side of the comparison is the trained transformer. If $S = T$, it measures whether the transformer behaves like exact ridge regression on the probes. If $S = T_Q$, it measures whether the transformer's local response is close to the reduced operator induced by an activation subspace. The denominator is the total ridge-response energy

on the same probe set, so the score is read as a relative root-mean-square response error. A small value means the discrepancy is small compared with the prediction changes that ridge regression itself would make on those directions.

The second quantity, $E(S_1, S_2; U)$, removes the transformer from the comparison and only asks whether two linear operators act similarly on the same probes. This is why later sections can separate transformer-to-reduced-operator error, $E_\varepsilon(F, T_Q)$, from subspace error, $E(T_Q, T)$.

Task-distribution probes test the directions emphasized by training; isotropic probes test a stronger global response. If $u \sim \mathcal{N}(0, I)$, then $\mathbb{E}\|Mu\|_2^2 = \|M\|_F^2$, so isotropic Gaussian probes estimate Frobenius operator error.

The operator errors above assume that the transformer has a stable local linear response around the observed label vector. A small transformer-to-ridge error is hard to interpret if the finite-difference response changes strongly when two probe directions are combined. The additivity defect is a guardrail for this issue: it checks whether the local response to $u + v$ is approximately the sum of the local responses to u and v . Because it is normalized by ridge-response energy, values below one mean that the non-additive interaction is smaller than the typical ridge response on the tested paired probes; values around one mean the interaction is ridge-scale; values above one mean the interaction is larger than the ridge response, so a local-linear operator interpretation is unreliable.

Definition 4 (Additivity defect). *For paired probe directions $V = \{(u_j, v_j)\}_{j=1}^m$, define*

$$C_j := F(y + \varepsilon u_j + \varepsilon v_j) - F(y + \varepsilon u_j - \varepsilon v_j) - F(y - \varepsilon u_j + \varepsilon v_j) + F(y - \varepsilon u_j - \varepsilon v_j),$$

$$A_\varepsilon(F; V) := \left(\frac{\sum_{j=1}^m \|C_j\|_2^2}{\sum_{j=1}^m \|T(\varepsilon u_j + \varepsilon v_j)\|_2^2 + \eta_{\text{num}}} \right)^{1/2}.$$

The four evaluations in C_j cancel exactly for an affine function of the labels. Thus a small $A_\varepsilon(F; V)$ says that, at the chosen scale and probe law, the transformer’s local response is close enough to linear that comparing it to a linear operator is meaningful. A large value would mean that the transformer may be nonlinear in the neighborhood being probed, so errors such as $E_\varepsilon(F, T)$ would mix two effects: failure to be ridge and failure to be locally linear. The additivity defect is only a validity check for the operator diagnostic; by itself it is not evidence that the linear operator is the ridge operator.

3.1 Reduced Operators from Activations

The behavioral test says whether the transformer response matches ridge regression, but not where the computation is represented. The next step is to ask whether the context-token activations define a small trial space from which ridge regression can be reconstructed. This is why the basis Q lives on the context-label side rather than in target space.

Definition 5 (A -orthonormal trial basis and reduced operator). *A matrix $Q \in \mathbb{R}^{n_{\text{ctx}} \times r}$ is A -orthonormal if $Q^\top A Q = I_r$. The reduced operator induced by Q is the linear operator*

$$T_Q := K_t Q Q^\top.$$

For a general full-rank basis $Z \in \mathbb{R}^{n_{\text{ctx}} \times r}$, the same operator is

$$T_Z := K_t Z (Z^\top A Z)^{-1} Z^\top.$$

The reduced operator is the restriction of ridge regression to the trial space. If $Q^\top A Q = I$, then the reduced dual vector

$$\alpha_Q = Q Q^\top y$$

is the A -orthogonal projection of $\alpha^\star = A^{-1}y$ onto $\text{col}(Q)$, and

$$T - T_Q = K_t(I - Q Q^\top A)A^{-1}.$$

Thus T_Q is not an arbitrary readout from activations. It is the ridge operator one obtains when the dual solution is forced to lie in the activation-derived context subspace.

Proposition 4 (Linear feature trial space). *For the linear kernel, let Z be any full-rank basis for $\text{col}(X_{\text{ctx}})$. Then the reduced operator*

$$T_Z = K_t Z (Z^\top A Z)^{-1} Z^\top$$

equals the exact operator $T = K_t A^{-1}$.

Proof. Fix y and write $\alpha^\star = A^{-1}y$ and

$$\alpha_Z = Z (Z^\top A Z)^{-1} Z^\top y.$$

The restricted normal equations give

$$Z^\top A (\alpha^\star - \alpha_Z) = 0.$$

Since $\text{col}(Z) = \text{col}(X_{\text{ctx}})$, this implies

$$X_{\text{ctx}}^\top A (\alpha^\star - \alpha_Z) = 0.$$

Let $r = \alpha^\star - \alpha_Z$. Using $A = X_{\text{ctx}} X_{\text{ctx}}^\top + \sigma^2 I$,

$$X_{\text{ctx}}^\top A r = (X_{\text{ctx}}^\top X_{\text{ctx}} + \sigma^2 I_{d_x}) X_{\text{ctx}}^\top r.$$

The matrix in parentheses is positive definite, so $X_{\text{ctx}}^\top r = 0$. Therefore $K_t r = X_{\text{tgt}} X_{\text{ctx}}^\top r = 0$, and $T_Z y = K_t \alpha_Z = K_t \alpha^\star = T y$ for every y . $\square$

Definition 6 (Frozen extraction map). *A frozen extraction map $\mathcal{E}$ maps the unperturbed episode and activations to an A -orthonormal basis*

$$Q = \mathcal{E}(G, y, \text{activations}).$$

After Q is extracted, the reduced operator T_Q is held fixed when label perturbations are tested.

Freezing Q is essential. Otherwise the basis could change with every probe direction, and the test would no longer certify one stable reduced operator carried by the unperturbed activations. In the experiments, this abstract map is instantiated in three concrete ways. Experiment 1 uses final-state weighted-SVD extractors: a raw extractor from the unperturbed final context hidden state, and a response-only extractor from final-state finite differences on build probes. Experiment 2 uses the layerwise activation-span extractor, which builds a basis from hidden-state finite-difference responses after removing directions already explained by earlier layers. Experiment 3 uses the raw final-state extractor from Experiment 1, because the causal intervention removes directions from the actual residual stream. In all cases the extraction uses only the unperturbed episode and build probes; evaluation is performed afterward on held-out probes with Q fixed.

Definition 7 (Pointwise errors). *For a frozen basis Q , define*

$$e_{\text{sub}}(Q; G, y) = \frac{\|T_Q y - T y\|_2}{\|T y\|_2 + \eta_{\text{num}}}, \quad \tau_{\text{read}}(Q; G, y) = \frac{\|F(y) - T_Q y\|_2}{\|T y\|_2 + \eta_{\text{num}}}.$$

Then

$$\frac{\|F(y) - T y\|_2}{\|T y\|_2 + \eta_{\text{num}}} \leq e_{\text{sub}}(Q; G, y) + \tau_{\text{read}}(Q; G, y).$$

This pointwise statement is only a sanity check. The operator statement is stronger.

Theorem 2 (Operator residual decomposition). *For every frozen A -orthonormal basis Q and probe set U ,*

$$D_\varepsilon F(y)[u] - T u = \underbrace{D_\varepsilon F(y)[u] - T_Q u}_{\text{transformer-reduced-operator residual}} + \underbrace{(T_Q - T)u}_{\text{subspace residual}}.$$

Consequently,

$$E_\varepsilon(F, T; U) \leq E_\varepsilon(F, T_Q; U) + E(T_Q, T; U).$$

The two terms answer different questions. The subspace residual $E(T_Q, T)$ asks whether the activation subspace is sufficient for the ridge operator. The transformer-reduced-operator residual $E_\varepsilon(F, T_Q)$ asks whether the trained transformer’s local response is close to the reduced operator induced by that subspace. These two residuals are the quantities used for the mechanism claim; auxiliary explained-fraction summaries are computed in the experiment logs but are not needed for the note’s argument.

3.2 Final-State Extractors

The final-state extractor is the simplest way to ask whether the computation has left a usable context-side subspace in the residual stream. The object we need for the reduced operator is a basis $Q \in \mathbb{R}^{n_{\text{ctx}} \times r}$: its columns are directions over context positions. The final context hidden state has exactly this context axis. If $H_L^{\text{ctx}} \in \mathbb{R}^{n_{\text{ctx}} \times D}$ is the final context-token hidden-state matrix, then each hidden coordinate gives one length- n_{ctx} pattern across the context tokens. The span of these patterns is the part of context-label space that is linearly visible in the final context representation.

The extractor measures this visible span and converts it into the kind of basis used by the reduced operator. For any matrix $M \in \mathbb{R}^{n_{\text{ctx}} \times p}$ whose columns are context-token patterns, compute

$$A^{1/2} M = U \Sigma V^\top.$$

We keep the left singular directions whose singular values are at least $\tau_{\text{sv}} \sigma_1$, up to the cap r_{max} , and set

$$Q(M) := A^{-1/2} U_{\text{kept}}.$$

This guarantees $Q(M)^\top A Q(M) = I$. The $A^{1/2}$ weighting is not a cosmetic choice: the ridge system measures dual directions using A , so the extractor ranks activation patterns by size in the same geometry in which $T_Q = K_t Q Q^\top$ is defined.

Experiment 1 uses two choices of M . The response-only choice is built from task-label build directions and then held fixed when we test both task-label and isotropic evaluation probes.

The raw extractor uses

$$M_{\text{raw}} = H_L^{\text{ctx}}, \quad Q_L^{\text{raw}} := Q(M_{\text{raw}}).$$

This asks a broad availability question: does the final context state contain some subspace from which the ridge operator can be reconstructed?

The response-only extractor uses hidden-state changes caused by label perturbations. For build probes $u \in U_{\text{build}}$, define

$$Z_L(u) = \frac{H_L^{\text{ctx}}(y + \varepsilon u) - H_L^{\text{ctx}}(y - \varepsilon u)}{2\varepsilon},$$

and stack these matrices as

$$M_{\text{resp}} = [Z_L(u_1), \dots, Z_L(u_m)], \quad Q_L^{\text{resp}} := Q(M_{\text{resp}}).$$

This asks a stricter question: do the label-sensitive final-state directions contain a subspace whose reduced operator is close to ridge regression?

After either basis is extracted, it is frozen. The experiment then computes the reduced operator $T_Q = K_t Q Q^\top$ and measures two held-out quantities: $E(T_Q, T)$, which tests whether the extracted subspace is sufficient for ridge, and $E_\varepsilon(F, T_Q)$, which tests whether the transformer’s local response is close to the reduced operator induced by that subspace.

For a bidirectional transformer, these bases may be target-conditioned. The claim is therefore about the conditional prediction operator $K_t A^{-1}$.

The final-state certificate is enough to test whether a suitable subspace is present at the end of the computation. It is not enough to say how that subspace arises across layers. The next section introduces a stricter diagnostic for that question.

4 Layerwise Activation Span

The final-state certificate says whether a useful subspace is present at the end of the computation. It does not say how such directions are distributed across the residual stream. This section defines a direct layerwise diagnostic: scan the context residual stream layer by layer, keep only directions that are label-sensitive, new relative to earlier accepted directions, and causally useful for the model response, then ask whether their span induces a reduced KRR operator close to T .

The diagnostic is deliberately post-training. It is not an architecture theorem. It measures what the trained transformer exposes in a particular checkpoint and episode. The reported dimension counts only directions that are directly accepted from residual-stream responses.

Fix build probes $U_{\text{build}} = \{u_1, \dots, u_m\}$. Let $H_\ell^{\text{ctx}}(y)$ be the context-token hidden-state matrix after layer ℓ , and define the finite-difference hidden response

$$Z_\ell(u) := \frac{H_\ell^{\text{ctx}}(y + \varepsilon u) - H_\ell^{\text{ctx}}(y - \varepsilon u)}{2\varepsilon}.$$

Stack the build-probe responses as

$$M_\ell = [Z_\ell(u_1), \dots, Z_\ell(u_m)].$$

The columns of M_ℓ are context-position patterns that are visible in layer ℓ when labels move. A direction that does not appear in these finite differences is not counted as evidence for a label-computation mechanism.

The construction accepts directions one layer at a time. If Q is an A -orthonormal basis of a context-position subspace, the A -orthogonal projection of a matrix M onto that subspace is

$$P_Q^A M := Q Q^\top A M.$$

Before processing layer ℓ , suppose that the accepted direction matrices $S_0, \dots, S_{\ell-1}$ have already been constructed. Each S_j has A -orthonormal columns and represents directions first accepted at layer j . The directions already available before layer ℓ are

$$R_\ell^{\text{pre}} := \text{span}\{S_j : 0 \leq j < \ell\}.$$

For $\ell = 0$ this space is empty. Orthogonality is still measured in the A -geometry, but the span itself is the direct seed span of previously accepted residual-stream directions.

Let Q_ℓ^{pre} be an A -orthonormal basis of R_ℓ^{pre} . Remove the part of the layer response that is already explained by this earlier space:

$$N_\ell := M_\ell - P_{Q_\ell^{\text{pre}}}^A M_\ell = \left(I - Q_\ell^{\text{pre}}(Q_\ell^{\text{pre}})^\top A\right) M_\ell.$$

Thus N_ℓ is the label-sensitive layer response after removing directions already accepted from earlier layers. This is an accounting step: later layers do not receive credit for re-exposing the same context-position direction.

Candidate directions for layer ℓ are obtained by an A -weighted SVD of N_ℓ . This means computing

$$A^{1/2}N_\ell = U_\ell \Sigma_\ell V_\ell^\top.$$

The left singular vectors with $\sigma_i \geq \tau_{\text{sv}}\sigma_1$ are the candidates. A declared selection rule then chooses an index set I_ℓ of these candidates, with $s_\ell = |I_\ell|$. The experiment below uses a causal selection rule: a candidate is accepted only if removing it from the context residual stream changes the transformer's local finite-difference response by a predeclared amount. If the accepted indices are I_ℓ , define

$$S_\ell := A^{-1/2}U_{\ell, I_\ell}.$$

Then $S_\ell^\top A S_\ell = I$. The A -weighting matters because the reduced operator and the ridge dual variables are measured in the A -geometry. This SVD is therefore principal-component extraction in the geometry relevant to the ridge system, not in the raw Euclidean geometry of context-token activations.

After the last layer L , define the seed layerwise activation span

$$R_{\text{seed}} := \text{span}\{S_\ell : 0 \leq \ell \leq L\}, \quad d_{\text{seed}} := \dim R_{\text{seed}}.$$

Let Q_{seed} be an A -orthonormal basis of R_{seed} . The reduced operator induced by the seed span is

$$T_{Q_{\text{seed}}} = K_t Q_{\text{seed}} Q_{\text{seed}}^\top.$$

The main layerwise certificate is the reduced-operator error $E(T_{Q_{\text{seed}}}, T)$. The dimension d_{seed} is only an accounting statistic. It says how many directly accepted context-position directions the checkpoint exposes, but small operator error requires those directions to align with the task-visible ridge directions. This distinction is important: a span may have enough dimension and still miss part of the ridge operator, and a full-dimensional span can make $T_Q = T$ trivially without implying that the transformer itself behaves like ridge.

As a sanity check, the implementation also reports held-out closure defects. For each layer k , define the partial span

$$R_k := \text{span}\{S_\ell : 0 \leq \ell \leq k\},$$

and let Q_k be an A -orthonormal basis of R_k . The closure defect is

$$\delta_k^{\text{close}} := \frac{\|(I - Q_k Q_k^\top A) M_k\|_{A,F}}{\|M_k\|_{A,F} + \eta_{\text{num}}}.$$

A small closure defect means that the accepted directions up to layer k can reconstruct the label-sensitive response observed at layer k . A large closure defect means that the final span may still work as an endpoint certificate, but the proposed layerwise account is missing much of the hidden response it claims to summarize.

Experiment 2 tests the following post-training hypothesis:

$$d_{\text{seed}}/r_{\text{task}} \approx 1 \implies E_{P_{\text{task}}}(T_{Q_{\text{seed}}}, T) \approx 0,$$

provided the selected directions align with the task-visible ridge directions. Agreement between the transformer response and the seed span is tested separately by $E_{\epsilon}(F, T_{Q_{\text{seed}}})$.

The high-level idea is now clear. The task-visible rank theorem says that ridge regression does not need all possible context-label directions: under task-label probes, it can be represented by a low-dimensional context-side space. The layerwise diagnostic asks whether the transformer exposes a comparable set of directly visible and causally useful context-position directions in its residual stream. The desired evidence is simultaneous: d_{seed} should be on the scale of r_{task} , $T_{Q_{\text{seed}}}$ should have small operator error, and $E_{\epsilon}(F, T_{Q_{\text{seed}}})$ should be close to $E_{\epsilon}(F, T)$. This combination means that the residual stream exposes a ridge-sufficient subspace and that the transformer locally responds according to the operator induced by that subspace.

At this point we have a behavioral test, an endpoint subspace certificate, and a layerwise activation-span diagnostic. The experiments now apply these objects in order. Experiment 1 tests behavior and the final-state certificate, including its low-rank alternative. Experiment 2 tests the layerwise span against the task-visible rank. Experiment 3 tests whether the directions that the reduced operator ranks as important are causally used.

5 Experiments

The experiments follow the certificate in order. The common design principle is to separate construction from evaluation: subspaces and fitted controls are built from one probe set, while numerical claims are reported on held-out label directions. This is the experimental analogue of freezing Q in the theory. The experiments are therefore not three independent demonstrations. They are a sequence of checks that close the gaps left by the theorem: behavior first, representation second, and causal use third.

5.1 Experimental Setup

Transformer checkpoint. Unless otherwise stated, we use checkpoint `model_L8.pt`: an 8-layer bidirectional transformer with $d_x = 5$, width $D = 128$, 4 attention heads, two-layer GELU FFNs, residual connections, no LayerNorm, no positional encodings, and a linear readout on target tokens.

Training distribution. Each training episode samples a fresh linear regression task. The input covariance is $\Sigma = U\Lambda U^\top$, with U drawn by QR-orthogonalizing a Gaussian matrix and Λ drawn uniformly from three spectral families: polynomial decay, exponential decay, and step spectra. Labels are $y = x^\top \beta + \epsilon$ with $\beta \sim \mathcal{N}(0, I)$ and $\epsilon \sim \mathcal{N}(0, \sigma^2)$, using $\sigma^2 = 0.1$. Training uses 50 points per episode with $n_{\text{tgt}} \sim \text{Unif}\{1, \dots, 10\}$ and $n_{\text{ctx}} = 50 - n_{\text{tgt}}$.

Evaluation geometry. The operator experiments use the linear kernel $K = XX^\top$ and $A = K + \sigma^2 I$. “Flat-spectrum OOD” means that the covariance eigenvalues are all equal to one random scale $c \sim \text{Unif}[1, 10]$; this spectrum is not present in training. Experiments 1–3 use $n_{\text{ctx}} = 47$ context points. Experiment 1 uses $n_{\text{tgt}} = 3$; Experiments 2–3 use $n_{\text{tgt}} = 16$.

Probe and extraction settings. All finite differences use $\varepsilon = 10^{-3}$ and numerical floor 10^{-12} . Weighted-SVD extraction uses $\tau_{\text{sv}} = 10^{-3}$. For a fixed episode geometry, task-label probes are sampled as $u \sim \mathcal{N}(0, A)$, implemented as $u = z \text{chol}(A)^\top$ with $z \sim \mathcal{N}(0, I_{n_{\text{ctx}}})$. Isotropic stress probes are sampled as $u \sim \mathcal{N}(0, I_{n_{\text{ctx}}})$. The interpolation sweep uses independent draws

$$u_\lambda = (1 - \lambda)^{1/2} u_{\text{task}} + \lambda^{1/2} u_{\text{iso}},$$

so its covariance is $(1 - \lambda)A + \lambda I_{n_{\text{ctx}}}$. Experiment 1 uses 4 episodes, 16 build probes, 32 held-out evaluation probes, 4 additivity probe pairs, and $r_{\text{max}} = L + 1 = 9$. Experiment 2 uses 16 episodes per setting, 16 build probes, 16 hidden-validation probes, 8 causal-selection probes, 16 held-out evaluation probes, causal damage threshold 0.05, and at most 16 SVD candidates per layer. It includes depth checkpoints $L \in \{2, 4, 6, 8, 12\}$, head checkpoints $H \in \{1, 2, 8\}$, and input-dimension checkpoints $d_x \in \{3, 5, 8, 10, 15\}$, plus an additional $d_x = 30$ stress checkpoint trained with the same 10k-step recipe. Experiment 3 uses 32 episodes, 32 causal probes, $r_{\text{max}} = 12$, and removes $k \in \{1, 2, 4\}$ directions. Auxiliary rank checks use 1024 sampled geometries per d_x at $(\epsilon, \alpha) = (0.01, 0.05)$. Reported random seeds are 123 for Experiment 1 and 42 for Experiments 2–3.

The setup deliberately uses flat-spectrum evaluation geometries for the operator experiments. This makes the rank requirement easy to interpret and keeps the geometry separate from the spectral curriculum used in training. Task-label probes then test the label directions induced by the sampled evaluation geometry. Experiment 1 also includes isotropic probes as a stricter stress test; Experiments 2–3 report task-label probes because their claim is task-local.

5.2 Experiment 1: Final-State Operator Certificate

Question. Does the transformer locally behave like exact KRR, and if so, can that behavior be explained through a subspace visible in the final context-token activation state? The first test is behavioral: on held-out label perturbations, is $D_\varepsilon F(y)$ close to the exact ridge operator T ? Only after that test passes does it make sense to ask whether the agreement factors through an activation-derived reduced operator.

For any frozen basis Q , the operator residual decomposes as

$$D_\varepsilon F(y)[u] - Tu = (D_\varepsilon F(y)[u] - T_Q u) + (T_Q - T)u.$$

Thus Experiment 1 is organized around three checks. First, the transformer should be close to exact KRR: $E_\varepsilon(F, T)$ should be small. Second, the reduced operator T_Q should be close to exact KRR. Third, the transformer should be close to T_Q . If only the first condition held, we would have behavior without a representation certificate. If only $T_Q \approx T$ held, the subspace would be sufficient for KRR but not tied to the transformer’s response. If only $F \approx T_Q$ held, the transformer and the subspace could agree on the wrong operator.

Procedure. For each episode, fix G and compute A , $T = K_t A^{-1}$, and Ty . Draw task-label build probes and held-out evaluation probes. Extract Q_L^{raw} and Q_L^{resp} . Here Q_L^{raw} is the basis extracted from the unperturbed final context-token hidden-state matrix H_L^{ctx} : it uses all context-side directions visible in the final activation state. Q_L^{resp} is extracted instead from finite-difference changes of that final hidden state under task-label build perturbations: it keeps only directions that move when the context labels are changed along directions generated by the task. Both bases are then frozen. We evaluate them on held-out task-label probes and on held-out isotropic probes; the isotropic rows are therefore a transfer test of the task-extracted bases, not a new isotropic extraction. For each basis, form the reduced operator $T_Q = K_t Q Q^\top$.

The comparison control is a same-span least-squares readout. For any perturbation u , both operators first form the same activation coordinates

$$z = Q^\top u.$$

The reduced KRR operator uses the KRR-implied readout from these coordinates:

$$T_Q u = (K_t Q)(Q^\top u), \quad C_{\text{KRR}} = K_t Q.$$

The control keeps the same features $Q^\top u$, but fits the readout directly to the transformer’s local finite-difference responses on build probes. If U_{build} is the build set, it solves

$$C_Q \in \arg \min_{C \in \mathbb{R}^{n_{\text{tgt}} \times r}} \sum_{u \in U_{\text{build}}} \|D_\varepsilon F(y)[u] - C Q^\top u\|_2^2, \quad T_{\text{actLR}} := C_Q Q^\top.$$

Thus, on a new perturbation u ,

$$T_{\text{actLR}} u = C_Q (Q^\top u).$$

The only difference is the readout from the same Q -features: T_Q uses the fixed KRR readout $K_t Q$, while T_{actLR} uses the fitted readout C_Q . The subscript actLR abbreviates “activation-span low-rank response fit.” This control answers a decodability question: using the same subspace coordinates, how well can a freely fitted target-side map reproduce the transformer’s local responses? It is not the KRR certificate, because its target-side coefficients are chosen from the transformer rather than from the KRR geometry. Then evaluate T , T_Q , and T_{actLR} on held-out probes. The reported comparison also includes random A -orthonormal subspaces of the same rank; these fail by one to two orders of magnitude on task probes and are used only to exclude the explanation that any subspace of the same dimension would suffice.

Results. The first check passes: the transformer is already close to exact KRR on the task-local perturbations we care about. On the sampled task labels, the transformer has MSE 0.0161 against the noiseless teacher targets, while exact KRR has MSE 0.0148; the ratio is 1.188. The pointwise relative transformer–KRR discrepancy is 0.01045. On held-out task-label perturbations, the finite-difference transformer–KRR error is $E_\varepsilon(F, T) = 0.0121$. This establishes the behavioral target that the rest of the experiment tries to explain through final activations. Additivity defects are much smaller than the operator errors: 4.4×10^{-4} on task probes and 3.5×10^{-3} on isotropic probes at $\varepsilon = 10^{-3}$, with the same scale at 10ε .

Table 1: Behavioral KRR check for the trained transformer, mean over 4 episodes. The operator error is measured on held-out task-label perturbations.

Quantity	Transformer	Exact KRR	Transformer–KRR
Pointwise MSE to teacher targets	0.0161	0.0148	–
Pointwise relative discrepancy	–	–	0.01045
Held-out operator error $E_\varepsilon(F, T)$	–	–	0.0121

We then ask whether this behavioral agreement can be explained through the final activation subspaces.

The raw final-state basis gives the cleanest certificate. On task probes, T_Q is very close to exact KRR ($E(T_Q, T) = 0.00148$), and the transformer is close to T_Q ($E(F, T_Q) = 0.01194$). The response-fitted control fits the transformer slightly better (0.00726), but it is much farther from

Table 2: Experiment 1 additivity defect, mean over 4 episodes. The defect is $(\sum_j \|D_\varepsilon F[u_j + v_j] - D_\varepsilon F[u_j] - D_\varepsilon F[v_j]\|^2 / (\sum_j \|T(u_j + v_j)\|^2 + \eta_{\text{num}}))^{1/2}$, estimated by the equivalent four-point central difference used in the code.

Probe distribution	$\varepsilon = 10^{-3}$	$\varepsilon = 10^{-2}$
Task-label probes $u, v \sim \mathcal{N}(0, A)$	4.43×10^{-4}	3.52×10^{-4}
Isotropic probes $u, v \sim \mathcal{N}(0, I)$	3.49×10^{-3}	2.93×10^{-3}

Table 3: Reduced operator vs. same-span response-fitted low-rank control, mean over 4 episodes. $E(F, S)$ denotes transformer-to- S operator error; $E(S, T)$ denotes S -to-ridge operator error.

Probe distribution	Basis	$E(F, T_Q)$	$E(F, T_{\text{actLR}})$	$E(T_Q, T)$	$E(T_{\text{actLR}}, T)$
Task probes	raw	0.01194	0.00726	0.00148	0.01202
Task probes	response	0.01239	0.01169	0.00367	0.01645
Isotropic probes	raw	0.2915	0.2007	0.05254	0.2687
Isotropic probes	response	0.3193	0.2418	0.1278	0.2929

exact KRR (0.01202). This does not make the control a failure; it shows that optimizing the readout for transformer fit is a different question from certifying that the same subspace supports the KRR operator. The response-only basis gives the same task-probe pattern, with T_Q still close to KRR (0.00367) and the response-fitted control farther from KRR (0.01645).

Random subspaces of the same ranks do not explain the result. On task probes, the raw random control has $E(T_Q, T) = 0.117$ and $E(F, T_Q) = 0.118$, roughly two orders of magnitude worse than the raw final-state basis. The response random control is similar, with $E(T_Q, T) = 0.0955$ and $E(F, T_Q) = 0.0955$.

The result is task-local. At the isotropic endpoint, the transformer-KRR operator error rises to about 0.302. For the raw basis, the reduced operator also degrades but remains closer to KRR than the transformer response: $E(T_Q, T) = 0.0525$ while $E(F, T_Q) = 0.292$. The task-built response-only basis also degrades off task: $E(T_Q, T) = 0.128$ and $E(F, T_Q) = 0.319$ on isotropic probes. Thus the response basis is useful as a task-local certificate, but it is not a global explanation of arbitrary label directions. The interpolation sweep in Figure 2 shows this transition from task-label directions to isotropic directions.

Interpretation. Experiment 1 gives the endpoint certificate used by the rest of the note. On task-label perturbations, the transformer is close to KRR, the raw final activation subspace induces a reduced operator close to KRR, and the transformer is close to that reduced operator. The response-fitted control shows that a better fitted readout from the same subspace is not automatically a better KRR explanation: it improves transformer fit slightly but moves away from KRR. The random-subspace controls show that the result is not a generic consequence of using a low-dimensional subspace of the same rank. The isotropic probes show the limit of the claim: Experiment 1 certifies a task-local operator explanation, not a global KRR implementation on all label directions. This experiment is therefore the necessary endpoint check: without it, the layerwise and causal experiments could only show that the transformer contains and uses some low-dimensional activation directions, not that those directions realize the task-local KRR operator we want to explain.

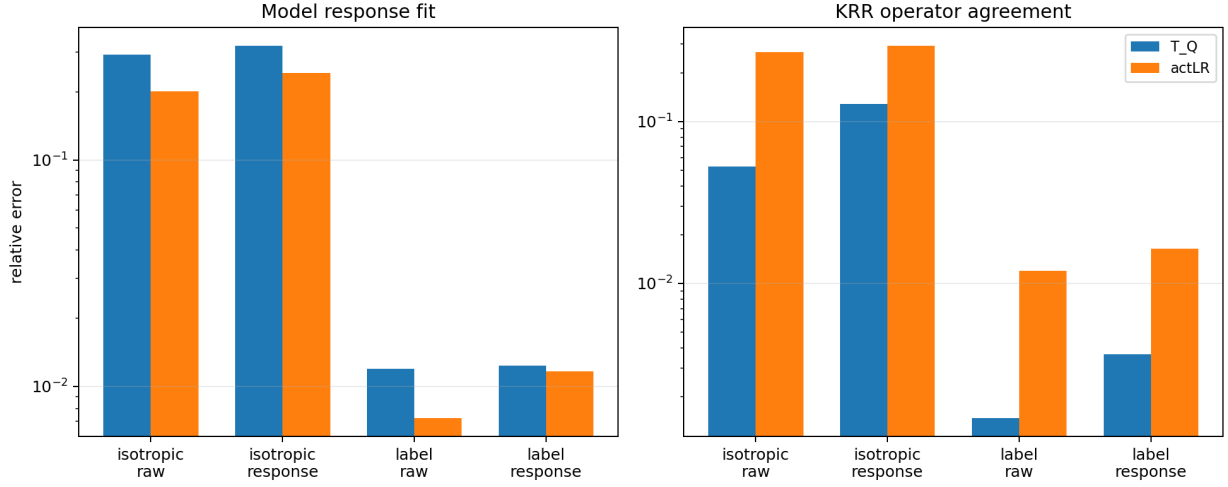


Figure 1: Reduced operator T_Q vs. the same-span response-fitted low-rank control T_{actLR} . Left panel: transformer-to-operator response error $E_\epsilon(F, S)$, with $S = T_Q$ or $S = T_{actLR}$. Right panel: operator-to-ridge error $E(S, T)$. On task probes, the response-fitted control fits the transformer slightly better but agrees substantially worse with the ridge operator.

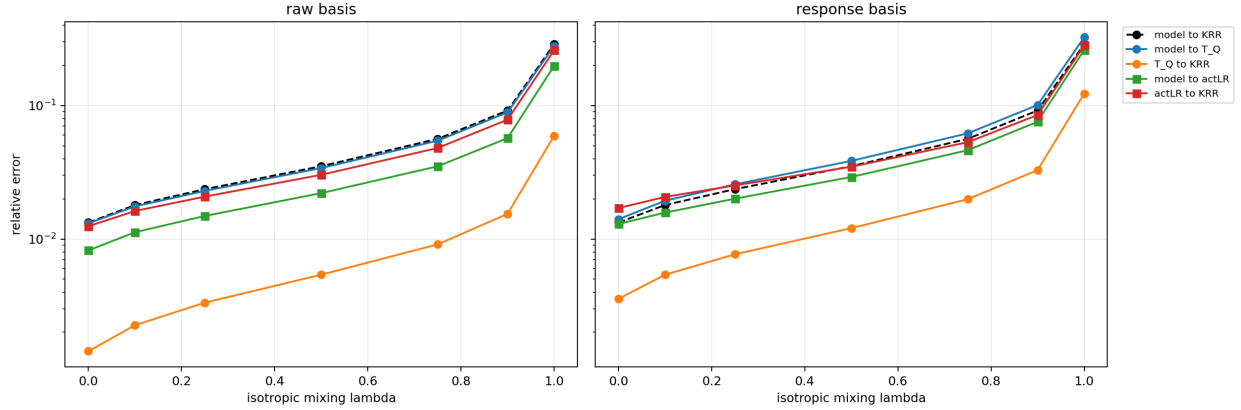


Figure 2: Interpolation sweep from task-label probes to isotropic probes. The horizontal axis is the probe mixing parameter λ , with $\lambda = 0$ equal to task probes and $\lambda = 1$ equal to isotropic probes. The plotted curves are $E_\epsilon(F, T)$, $E_\epsilon(F, T_Q)$, $E(T_Q, T)$, $E_\epsilon(F, T_{actLR})$, and $E(T_{actLR}, T)$.

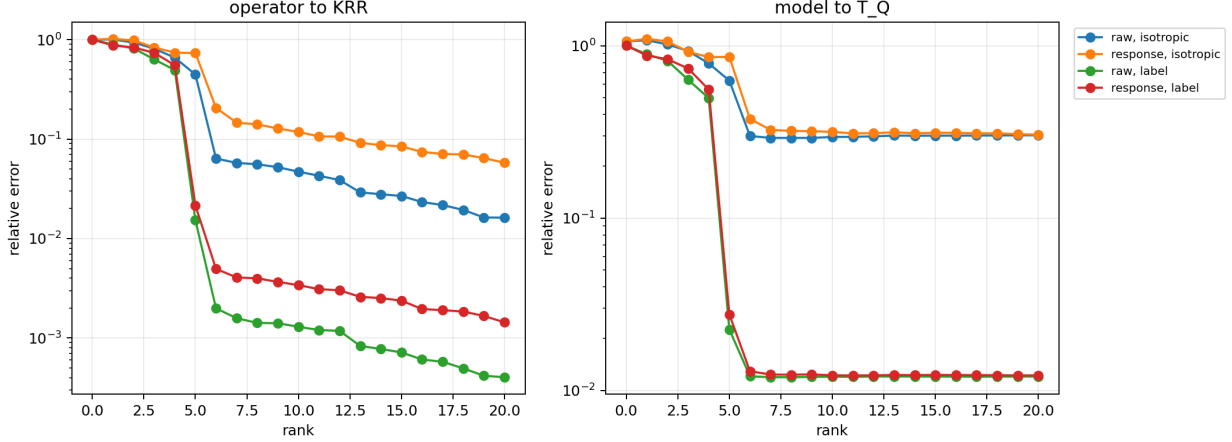


Figure 3: Rank curves for the raw and response-only bases. The horizontal axis is the retained basis rank r . Left panel: reduced-operator error $E(T_Q, T)$. Right panel: transformer-to-reduced-operator response error $E_\varepsilon(F, T_Q)$. Curves are shown separately for task-label and isotropic probes.

5.3 Experiment 2: Causal Layerwise Rank and Task Rank

Question. Does the layerwise activation span contain the task-visible directions needed for ridge regression? Experiment 1 checked the endpoint: a final activation subspace can induce a reduced operator close to exact KRR. Experiment 2 asks how this subspace is supplied across layers. In the linear task, the required context-label dimension is the task-visible rank r_T , which agrees with d_x in the target-rank sweep up to the target-count cap. The question is therefore concrete: after we keep only directions that are both visible in layerwise label responses and causally useful for the transformer response, does their accumulated seed span have task-visible scale, and does the reduced operator induced by that span actually approximate T ?

Procedure. For each episode, fix the sampled geometry and compute $A = K + \sigma^2 I$ and $T = K_t A^{-1}$. Draw four independent task-label probe sets: build probes, hidden-validation probes, causal-selection probes, and held-out evaluation probes. The build probes propose candidate directions, the causal-selection probes decide which candidates are accepted, and the hidden-validation probes report closure diagnostics. The evaluation probes are used only after the span is frozen.

At layer ℓ , compute hidden-response matrices M_ℓ^{build} and M_ℓ^{val} from finite differences of the context-token hidden state. Suppose earlier layers have already selected directions whose accumulated span before layer ℓ has A -orthonormal basis Q_ℓ^{pre} . Remove the part already explained by this earlier span:

$$N_\ell^{\text{build}} = \left(I - Q_\ell^{\text{pre}} (Q_\ell^{\text{pre}})^\top A \right) M_\ell^{\text{build}}, \quad N_\ell^{\text{val}} = \left(I - Q_\ell^{\text{pre}} (Q_\ell^{\text{pre}})^\top A \right) M_\ell^{\text{val}}.$$

Thus N_ℓ^{build} is the novel label-sensitive layer response: it is what remains after earlier accepted directions have been accounted for using the same A -orthogonality used by the reduced operator.

Candidate directions are obtained from the A -weighted SVD

$$A^{1/2} N_\ell^{\text{build}} = U_\ell \Sigma_\ell V_\ell^\top, \quad q_i = A^{-1/2} U_{\ell, i}.$$

The run keeps at most 16 candidates with singular value at least $10^{-3} \sigma_1$. These candidates are not automatically accepted. For each candidate q_i , perform a causal removal after layer ℓ : replace the

context hidden state H_ℓ^{ctx} by

$$H_\ell^{\text{ctx}} - q_i q_i^\top A H_\ell^{\text{ctx}},$$

roll the remaining layers forward, and measure the finite-difference response damage on the causal selection probes:

$$\Delta_\ell(q_i) = \frac{\|D_\varepsilon F_{\ell, q_i}^{\text{rem}}(y) - D_\varepsilon F(y)\|_F}{\|D_\varepsilon F(y)\|_F + \eta_{\text{num}}}.$$

Here $F_{\ell, q_i}^{\text{rem}}$ denotes the transformer after removing direction q_i from the context residual stream at layer ℓ . A candidate is accepted when $\Delta_\ell(q_i) \geq 0.05$. The accepted directions at all layers define Q_{seed} , an A -orthonormal basis of $\text{span}\{S_0, \dots, S_L\}$. Finally, freeze Q_{seed} , form $T_{Q_{\text{seed}}} = K_t Q_{\text{seed}} Q_{\text{seed}}^\top$, and evaluate $E(T_{Q_{\text{seed}}}, T)$ and $E_\varepsilon(F, T_{Q_{\text{seed}}})$ on held-out task-label probes. The point is not that d_{seed} must equal r_T exactly. The seed span counts directions used by the trained transformer, whereas r_T is the optimal ridge rank for task-distributed label perturbations. The decisive quantity is the operator error $E(T_{Q_{\text{seed}}}, T)$.

Results. The 16-episode target-rank sweep uses the causal selection rule described above. Let r_T be the tail-based empirical task-visible rank of the ridge operator at RMS tolerance $\epsilon = 0.01$, and let $d_{\text{seed}} = \dim R_{\text{seed}}$. The results are:

Table 4: Experiment 2 target-rank sweep with causal seed-span selection.

d_x	r_T	d_{seed}	d_{seed}/r_T	$E(T_{Q_{\text{seed}}}, T)$	$E(F, T)$	$E(F, T_{Q_{\text{seed}}})$
3	3.0	4.6	1.542	0.0011	0.0063	0.0064
5	5.0	6.0	1.200	0.0014	0.0091	0.0093
8	8.0	9.0	1.125	0.0017	0.0129	0.0132
10	10.0	11.0	1.100	0.0014	0.0131	0.0131
15	14.9	15.8	1.063	0.0849	0.0174	0.0960
30	16.0	47.0	2.938	0.0000	1.6867	1.6867

For $d_x = 3, 5, 8, 10$, the direct seed span contains slightly more directions than the task-visible rank and the reduced-operator error is about 10^{-3} . This is the desired pattern for the layerwise diagnostic: the model exposes a small number of directly accepted residual-stream directions, those directions are on the same scale as the ridge rank requirement, and restricting ridge to their span loses essentially none of the task-local operator response.

For $d_x = 15$, the selected seed dimension is not short on average ($d_{\text{seed}} = 15.8$, $r_T = 14.9$), but the reduced-operator error rises to 0.0849. The failure is therefore an alignment failure, not merely a rank shortage: the accepted directions have enough dimension on average, but they do not consistently span all task-visible ridge directions.

The additional $d_x = 30$ row is a stress check rather than a clean continuation of the earlier rank sweep. Because this experiment uses $n_{\text{tgt}} = 16$, the task-visible rank is capped at 16 even though the input dimension is 30. In the seed-span rerun, causal selection expands to the full context-position space, $d_{\text{seed}} = 47$, so the reduced operator is exactly KRR:

$$E(T_{Q_{\text{seed}}}, T) = 0.$$

But the transformer itself is far from ridge:

$$E(F, T) = 1.6873.$$

Thus $d_x = 30$ should be read as a high-dimensional undertrained/capacity stress case. A full seed span can make $T_Q = T$ trivially, but this does not imply that the model locally behaves like KRR. The model-subspace alignment error catches the failure. This row is useful precisely because it prevents an overreading of $E(T_Q, T)$: representation sufficiency alone is not a mechanism claim unless the transformer’s local response also follows the induced operator.

Table 5: Experiment 2 fixed- $d_x = 5$ depth/head sweep with causal seed-span selection, mean over 16 episodes per checkpoint. All checkpoints use width $D = 128$.

Checkpoint	L	H	d_{seed}	r_T	$E(T_Q, T)$	$E(F, T_Q)$	$E(F, T)$
$L = 2$	2	4	16.3	5.0	0.00095	0.18537	0.18535
$L = 4$	4	4	6.2	5.0	0.00127	0.02534	0.02528
$L = 6$	6	4	6.1	5.0	0.00137	0.01272	0.01255
$L = 8$	8	4	6.0	5.0	0.00147	0.00962	0.00952
$L = 12$	12	4	6.1	5.0	0.00110	0.00953	0.00943
$H = 1$	8	1	6.2	5.0	0.00085	0.01558	0.01559
$H = 2$	8	2	6.1	5.0	0.00108	0.00908	0.00904
$H = 8$	8	8	6.0	5.0	0.00164	0.00996	0.00982

The depth/head sweep gives the necessary qualification. This is a fixed- d_x sweep: all these checkpoints are trained and evaluated at $d_x = 5$, with width $D = 128$, while depth varies over $L \in \{2, 4, 6, 8, 12\}$ at four heads and head count varies over $H \in \{1, 2, 4, 8\}$ at eight layers. Causal seed selection gives $E(T_{Q_{\text{seed}}}, T) \approx 10^{-3}$ for all reported checkpoints, but transformer accuracy still varies. The $L = 2$ checkpoint has $d_{\text{seed}} = 16.3$ and a strong reduced-operator certificate, yet $E(F, T) = 0.185$. By contrast, the $L = 8$ checkpoint has $d_{\text{seed}} = 6.0$, reduced-operator error 0.00147, and $E(F, T) = 0.0095$; the $H = 2$ and $L = 12$ checkpoints are similar, with transformer errors 0.0090 and 0.0094. The fixed- d_x sweep therefore separates two facts: the activations can expose a subspace from which KRR is recoverable, while the trained transformer may still fail to use that available subspace accurately. The seed span certifies availability of a ridge-sufficient operator subspace, not automatic behavioral KRR agreement.

Interpretation. Experiment 2 supports a resource law with an alignment condition:

$$d_{\text{seed}}/r_T \approx 1 \implies E_{P_{\text{task}}}(T_{Q_{\text{seed}}}, T) \approx 0.$$

The law should be read qualitatively, not as an exact rank identity. The seed span counts directions that are visible and causally consequential in the trained transformer, so it can include auxiliary directions that are not part of the optimal ridge rank. The task-visible rank is the lower-rank operator benchmark. The useful conclusion is that successful checkpoints expose a compact span on that scale, and that this span induces a small-error reduced ridge operator when it is aligned with the task-visible directions. It is not a statement about raw architecture counts, and it is not by itself a statement about transformer accuracy. This experiment was needed because Experiment 1 only showed that a suitable subspace is present at the final state; it did not show whether the layerwise label responses supply the task-visible number of directions. Experiment 2 shows that, for the linear task, the causally selected seed span has the task-visible scale in successful target-rank settings, and that operator error can still appear when the seed span has enough dimension but is not aligned with the task-visible ridge directions.

5.4 Experiment 3: Causal Direction Removal

Question. Are the task-important directions of the reduced operator causally used by the transformer? This is needed because a direction can be decodable without being used. The surgery test removes directions ranked important by the reduced operator and checks whether the transformer’s task-local response changes accordingly.

Procedure. For an extracted raw final A -orthonormal basis $Q = [q_1, \dots, q_r]$, define Q_{-j} by removing column j from Q . Direction q_j is important for the reduced operator if deleting it changes the reduced operator’s response. On a probe set U , measure this by

$$\Lambda_j^{\text{red}}(U) := \left(\frac{\sum_{u \in U} \|(T_Q - T_{Q_{-j}})u\|_2^2}{\sum_{u \in U} \|Tu\|_2^2 + \eta_{\text{num}}} \right)^{1/2}.$$

Task leverage means this score computed on task-label probes; isotropic leverage means the same score computed on isotropic probes. The main intervention set consists of the top- k directions by task leverage.

We intervene on the context-token residual state after residual state $s = 0, \dots, L$ by replacing

$$H_s^{\text{ctx}} \mapsto H_s^{\text{ctx}} - \sum_{j \in J} q_j q_j^\top A H_s^{\text{ctx}},$$

then roll the remaining layers forward. The intervention removes the A -orthogonal projection of the context residual stream onto directions that the reduced operator predicts to be important. Matched controls remove low-task-leverage directions from the same extracted basis, high-isotropic-but-low-task directions, high-activation-variance-but-low-task directions, and random subsets of Q . The low-task directions are the cleanest negative control because they share the same extraction pipeline and residual-stream coordinates.

Results. Before surgery the checkpoint is in the task-local reduced-operator regime:

$$E_{\text{task}}(F, T) = 0.00926, \quad E_{\text{task}}(F, T_Q) = 0.00920, \quad E_{\text{task}}(T_Q, T) = 0.00086.$$

The same episodes retain the off-distribution response drift seen above:

$$E_{\text{iso}}(F, T) = 0.266, \quad E_{\text{iso}}(T_Q, T) = 0.0378.$$

Surgery is therefore applied in a regime where the task-local certificate holds before intervention. At residual state $s = 1$, removing a single direction predicted to matter most for the reduced operator causes prediction drift $4.68\times$, KRR-error inflation $649\times$, and task response damage 41.59 . Removing a single low-task-leverage direction from the same basis is nearly inert: prediction drift $9.8 \cdot 10^{-4}$, KRR-error inflation $1.03\times$, and task response damage 0.0024 .

The layer sweep localizes the effect. High-task surgery is most damaging at early residual states, before later attention has routed the relevant context information into target tokens. Context-only surgery after the final layer is a no-op, as expected from the architecture. At $s = 1$, the relationship between aggregate task leverage and task response damage is strong across matched non-random controls (log-log Pearson 0.91 , Spearman 0.87). This is the key causal specificity check: damage tracks task leverage, not merely activation energy or membership in the extracted subspace.

As a complementary mediation check, we also intervene on the whole extracted subspace at residual state $s = 1$. Keeping only the A -projection of the context residual stream onto Q ,

$$H_s^{\text{ctx}} \mapsto QQ^\top AH_s^{\text{ctx}},$$

nearly preserves the task-local response: over 8 episodes,

$$E_{\text{task}}(F, T) = 0.01866, \quad \text{response damage} = 0.01659.$$

Removing the same subspace,

$$H_s^{\text{ctx}} \mapsto H_s^{\text{ctx}} - QQ^\top AH_s^{\text{ctx}},$$

destroys the response:

$$E_{\text{task}}(F, T) = 145.43, \quad \text{response damage} = 145.43.$$

Thus, in this intervention family, the complement of Q is largely dispensable for the task-local KRR response, while Q itself is necessary.

Whole-subspace intervention at residual state $s = 1$

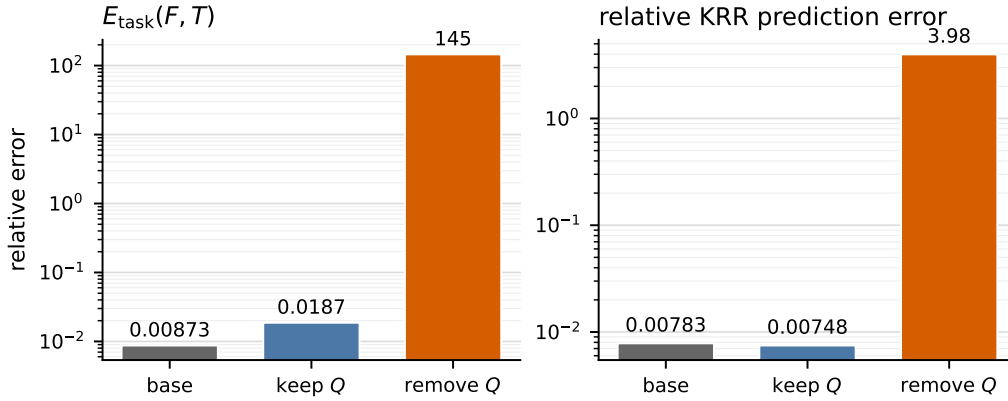


Figure 4: Experiment 3: whole-subspace complement ablation at residual state $s = 1$. Left panel: task finite-difference response error $E_{\text{task}}(F, T)$. Right panel: relative KRR prediction error $\|F_{\text{surg}}(y) - Ty\|_2 / \|Ty\|_2$. Keeping only the A -projection onto Q nearly preserves both quantities, while removing Q destroys both.

Interpretation. The directions that the reduced operator identifies as task-relevant are not only decodable and sufficient for small reduced-operator error; removing them before later attention can use them strongly damages both prediction and the task-local finite-difference response.

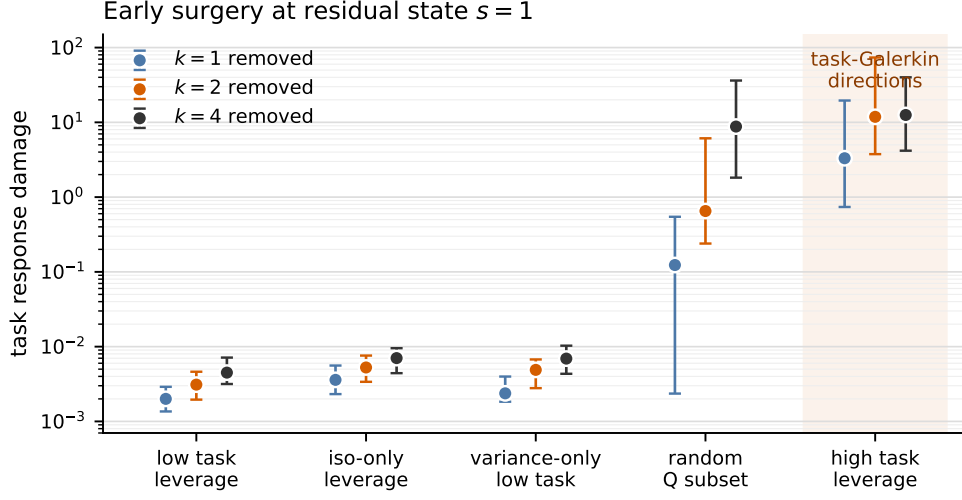


Figure 5: Experiment 3: causal effect of task-important activation directions. The plotted metric is task finite-difference response damage $(\sum_j \|D_\epsilon F_{\text{surg}}(y)[u_j] - D_\epsilon F_{\text{base}}(y)[u_j]\|^2 / (\sum_j \|Tu_j\|^2 + \eta_{\text{num}}))^{1/2}$, for task probes $u_j \sim \mathcal{N}(0, A)$. High task-leverage directions produce orders-of-magnitude larger damage than low-task-leverage directions from the same activation basis.

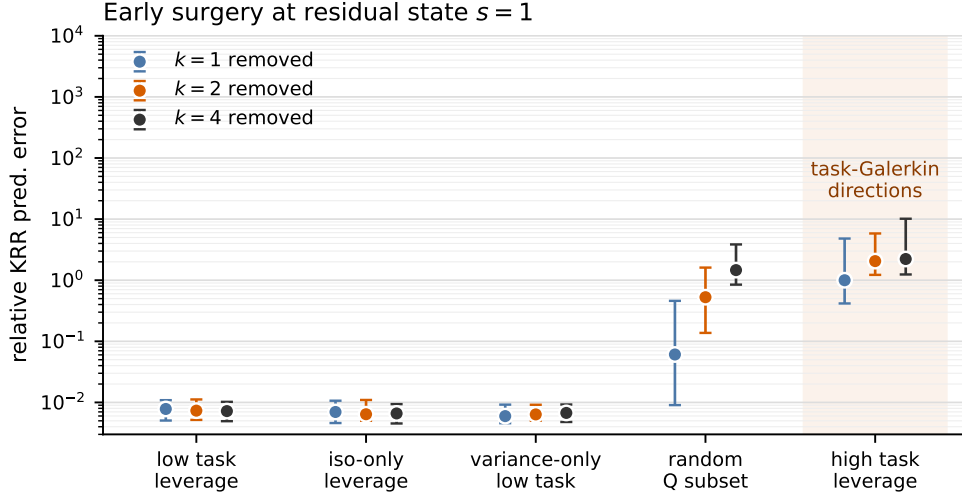


Figure 6: Experiment 3: matched surgery plot for relative KRR prediction error $\|F_{\text{surg}}(y) - Ty\|_2 / \|Ty\|_2$. The same residual state, controls, and removed direction counts as Figure 5 are used. High task-leverage directions also produce large ordinary prediction degradation, while low-task-leverage directions remain nearly inert.

6 Falsification and Conclusion

The expected observation would be only that a transformer trained on linear regression gives KRR-like predictions. That is not the main claim here. Pointwise prediction accuracy alone does not identify the computation inside an in-context learner: many rules can agree on the sampled labels while having a different response to nearby label perturbations.

The stronger claim is operator-level and task-local. After fixing the input geometry, KRR is the linear map $T = K_t A^{-1}$ from context labels to target predictions. We therefore ask whether the transformer has the same local label-to-prediction response, whether that response can be explained through a compact activation subspace, and whether the relevant directions in that subspace are causally used.

The evidence has three parts. First, on held-out task-label perturbations, the transformer’s finite-difference response is close to the exact KRR operator. Second, the activation-derived seed span is on the task-visible scale in successful settings and its induced reduced operator is close to exact KRR. The task-visible rank is computed before looking at activations; it is the number of directions KRR needs under the task label distribution. Third, the high-leverage directions of the reduced operator are not merely decodable: removing them from the residual stream strongly damages prediction and the task-local response, while matched low-leverage removals are nearly inert.

These observations make the result more specific than a generic low-rank explanation. The same-span response-fitted low-rank control can fit the transformer’s response slightly better on task probes, but it agrees much worse with KRR. The reduced operator used here is constrained by the KRR system itself, so success means that the extracted subspace supports the ridge computation, not just that some low-rank response fit works after the fact.

The claim is also not global. Isotropic label probes show clear response drift: the transformer has not learned a uniformly correct inverse on all possible label directions. The supported conclusion is narrower and more mechanistic. On the label directions emphasized by the training distribution, the trained transformer exposes a compact context-side activation subspace whose induced KRR operator is accurate, whose dimension is on the task-visible scale in successful settings, and whose high-leverage directions are causally important for the transformer’s response.

The account is falsifiable at each step. If $E_\epsilon(F, T)$ is large while pointwise error is small, the transformer has not learned the conditional KRR response. If $E(T_Q, T)$ is large, the extracted subspace does not explain the KRR operator. If $E(T_Q, T)$ is small but $E_\epsilon(F, T_Q)$ is large, the subspace is sufficient but the transformer does not use it to produce its local response. If T_{actLR} explains held-out responses much better than T_Q while also staying close to KRR, the reduced-operator construction is not the right explanation. If leverage does not predict causal damage, the directions may be decodable but not causally operative.